

Offline Phase

DS1 offline training
Tiny 1-D CNN
(backbone + head)

Export
weights +
episodes

On-Device Phase (PSoC 6 CM4)

Frozen backbone
PSoC 6 CM4
emb. dim = 32

Linear head
SGD on W, b

Prototype head
class means p_c

DS2 support beats consumed only on-device

